Bilal Ahmed

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RESEARCH INTERESTS

Probabilistic Generative Modeling, Computational Imaging, Inverse Problems, Principled Learning Approaches for Low-Data Regimes, Learning Useful Latent (Quantized) Representations Using Principled Reconstruction Losses.

EDUCATION

Purdue University, PhD candidate — Aug. 2021 – Aug. 2026 (Expected)
Elmore Family School of Electrical and Computer Engineering. — GPA: 3.88/4.0
Advisor: Dr. Joseph G. Makin

University of Engineering & Technology Lahore — Aug. 2011 – Jun. 2015
B.Sc. in Electrical Engineering (Rank: 3/240) — GPA: 3.85/4.0

PROJECTS

Probabilistic Generative Modeling for Neural Speech Synthesis (in-progress)
Synthesizing speech from a small amount of neural recordings is a challenging problem. I am working on solving this using principled probabilistic approaches. One such promising approach is the use of normalizing flows to directly predict mel-spectrograms from stereo-EEG recordings. This works much better than using a discriminative model for the same problem. I am also working on cross-subject transfer and comparisons with other generative modeling approaches, such as inverse problem formulations.

Tuning Preferences of Neurons in the Auditory Cortex Using a Diffusion Prior (in-progress)
Using the inverse problem framework, we can understand the tuning preferences of neurons, provided that we have accurate encoding models. We approach this problem using DNNs as encoding models and diffusion models as learned priors. We generate synthetic stimuli through posterior sampling to elicit strong responses from auditory cortex neurons. We are collaborating with a non-human primate lab to test these synthetic stimuli in a closed-loop setup. This enables validation of auditory cortex encoding models and helps reveal neuronal tuning preferences.

Restart Posterior Sampling (RePS) for Solving Inverse Problems
Proposed Restart Posterior Sampling (RePS), an efficient method for solving inverse problems with diffusion priors. We evaluated our method on a standard set of five linear and three nonlinear inverse problems for image restoration tasks. The algorithm interleaves conditioned ODEs with intermittent restarts, benefiting from the low discretization error of ODEs and the error contraction induced by stochasticity. Demonstrated faster convergence and higher-quality samples than state-of-the-art methods.

Modeling the Auditory Cortex Using Speech Recognition Models
Studied deep neural networks trained for ASR as candidate encoding models of the auditory cortex. Analyzed six models varying in architecture and training objectives, and demonstrated that task-optimized DNNs predict neural spiking activity more accurately than traditional spectro-temporal filters.

PUBLICATIONS/PREPRINTS

Solving Diffusion Inverse Problems with Restart Posterior Sampling
Bilal Ahmed, Joseph G. Makin
arXiv (submitted for review)

Deep Neural Networks Explain Spiking Activity in Auditory Cortex
Bilal Ahmed, Joshua D. Downer, Brian J. Malone, Joseph G. Makin
PLOS Computational Biology, 2025.

ACADEMIC EXPERIENCE

Research Assistant Aug. 2025 – Present
Advisor: Dr. Joseph G. Makin
Purdue University

Teaching Assistant, Generative Models, (Graduate Course) Fall 2025
Instructor: Dr. Joseph G. Makin
Purdue University

Teaching Assistant, Generative Models, (Graduate Course) Spring 2024
Instructor: Dr. Joseph G. Makin
Purdue University

Research Intern Jul. 2014 – Sep. 2014
Manager: Dr. Waqar Mahmood
Al-Khawarizmi Institute of Comupter Science, UET Lahore, Pakistan

PROFESSIONAL LEADERSHIP EXPERIENCE

Instrumentation & Control Engineer Sep. 2015 – Jul. 2021
Fauji Fertilizer Company Limited, Pakistan

- Led a team of up to six technicians responsible for maintenance and reliability of real-time monitoring instrumentation and control systems in a large-scale chemical plant.
- Owned configuration and modification of the Distributed Control System (DCS), including implementing new control loops and deploying additional process monitoring.
- Led major reliability upgrades and commissioning projects (steam turbine governor replacement, vibration monitoring upgrade, chlorine dosing system), contributing to **zero plant shutdowns due to I&C failures for three consecutive years**.

Skills

Programming & Tools	Python, PyTorch, C/C++, MATLAB, Linux, LaTeX
ML Training	Torch Distributed, PyTorch Lightning, Hydra, Weights & Biases, TensorBoard, Optuna, HuggingFace
Expertise	Generative Models (Diffusion, Flow), Speech Models (ASR, TTS), Computational Imaging, Computer Vision